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* IN THE HIGH COURT OF DELHI AT NEW DELHI

Date of decision: 11th August, 2023

+ C.A.(COMM.IPD-PAT) 358/2022

MICROSOFT TECHNOLOGY LICENSING, LLC Appellant

Through: Ms. Vindhya S. Mani, Mr. Gursimran Singh Narula and Ms. Vaishali Joshi, Advocates.

versus

THE ASSISTANT CONTROLLER OF PATENTS AND DESIGNS

..... Respondent

Through: Ms. Arunima Dwivedi, CGSC with Mr. Aakash Pathak and Ms. Pinky Pawar, Advocates with Mr. Rakesh Khushwah, Patent Controller.

CORAM:

HON'BLE MR. JUSTICE SANJEEV NARULA

JUDGMENT

SANJEEV NARULA, J. (Oral)

1. Microsoft Technology Licensing, LLC, the Appellant, filed the Indian Patent Application No. 8360/DELNP/2010 on 24th November, 2010 for invention titled “System for Advanced Bi-directional Predictive Coding of Interlaced Video” [*hereinafter*, “*subject invention*”]. This application was stylised as a divisional application, stemming from the parent Indian Patent Application No. 487/DELNP/2010 dated 30th January, 2006 [*hereinafter*, “*parent application*”]. The subject invention met rejection from Assistant Controller of Patents and Designs on the ground of Sections 16(1), 16(3) and 3(k) of the Patents Act, 1970 *vide* order dated 28th November, 2019 [referred



to as “*impugned order*”]. Aggrieved, Appellant has preferred the present appeal under Section 117A of the Patents Act.

Subject invention and the divisional application

2. The subject invention pertains to a video compression technique. It is a system of encoding and decoding a current direct-mode macroblock in a current interlaced bi-directionally predicted field, in order to reduce the bit rate for transmission. Traditional video compression methodologies tend to consume significant memory owing to the coding of macroblocks across one to four motion vector modes. The Appellant posits that this technical quandary can be effectively addressed *via* the subject invention. This innovation minimizes memory utilization by judiciously selecting between motion vector modes and buffering the chosen field motion vectors, ultimately aiding in the prediction of a subsequent frame through bi-directional predictive coding.

3. Originally, the Appellant submitted the parent application encompassing 183 claims. However, following certain objections highlighted in the First Examination Report [“FER”] dated 25th February 2010, claims 155-183 were excised from the parent application on 07th December, 2010. The Appellant thereafter initiated a divisional application for the subject invention comprising of 29 claims.

4. Upon examination of the divisional application, another FER was generated on 11th September, 2017. This report underscored that the divisional application did not introduce a unique invention. Instead, its content echoed claims 155-183 of the parent application, which, by that date, had been declined. Consequently, the application failed to meet the



criteria for a divisional application as stipulated in Section 16 of the Patents Act.

5. On 09th February, 2018, the Appellant sought to address the concerns raised in the FER. Subsequently, a hearing notice was dispatched, scheduling a hearing on 17th September, 2019 to address the lingering objections, specifically under Sections 3(k) and 16 of the Patents Act. While the Appellant participated in this hearing, the Controller remained unpersuaded. Consequently, the impugned order was issued under Section 15 of the Patents Act, leading to the refusal of the Appellant's divisional application.

6. In the meantime, the order rejecting the parent application was set aside by this Court *vide* judgement dated 27th March, 2023 in C.A. (COMM.IPD-PAT) 106/2022. Consequently, upon fresh scrutiny, the parent application has been granted a patent on 25th June, 2023.

The impugned order

7. Relevant portion of the impugned order relating to Section 16 of the Patents Act, reads as under:

- “5. It is evident that the applicant had filed 1-183 claims in the First application (Parent application). There was multiple independent set of claims related to only 'A Method and A Computer readable medium storing computer executable instructions'. There was not even a single claim of 'A computing Device or A System for implementing the method claimed.
6. The applicant filed the current application with 1-29 claims by dividing the parent application of his own free will under section 16(1) of The Patent Act. No objection of distinct invention was communicated in FER of parent application from the office (Indian Patent Office).
7. The agent of the applicant has contested in his submission dated 03/10/2019 'the claims of the present application were dropped from the claim set originally filed for the parent application 487/DELNP/2006. It is pertinent to note that the parent application originally had 183 claims. Of the 183 claims, claims 155-183 were been deleted from the parent application at the time of responding to FER dated February 25, 2010. with the response being



filed on December 07, 2010. The claims 155-183 of the parent application formed the basis of the of the current divisional application'. Therefore, the applicant filed claims 1-29 in this current application (divisional application, but it is clear that these claims are not related to distinct invention.

8. *Further, the applicant has amended the 'method' claims 1-29 to 'system claims' 1-29. These amendments cannot be considered under section 16 as evidenced above.*
9. *In view of the above reasoning, I am of the opinion that the claimed matter in claims 1-29 is not a distinct invention under section 16 of The Patent Act."*

Counsel's submissions

8. Ms. Vindhya S. Mani, representing the Appellant, meticulously walks the Court through a detailed, claim-by-claim juxtaposition of the independent claims present in both the parent and divisional applications, aiming to illustrate the presence of multiple distinct inventions within the parent application. While acknowledging the precedent set by this Court in ***Boehringer Ingelheim International GMBH v. The Controller of Patents and Anr.***,¹ she contends that the Appellant's application is liable to succeed. This is because the subject of the divisional application has neither been encapsulated within the claims nor detailed in the specifications of the parent application. Given the distinct nature of the divisional application's claims, she asserts that the Appellant rightfully qualifies for the provisions under Section 16 of the Patents Act.

9. Ms. Arunima Dwivedi, representing the Respondent, defends the impugned order by relying on a comparison between the claims in the parent and the divisional applications and the jurisprudence to bolster her assertions. She emphasizes the lack of distinction between the claims 155-183 of the parent application and the 29 claims outlined in the divisional

¹ C.A.(COMM.IPD-PAT) 295/2022.



application. Anchoring her argument on this Court's ruling in the case of *Boehringer* (*Supra*). Ms. Dwivedi stresses on the necessity for “plurality of inventions” within the parent application's claims. She argues that in the case at hand, while the parent application encompasses multiple independent sets of claims, all centre around a method and a computer-readable medium containing executable instructions. Crucially, these independent sets converge towards a group of inventions, which collectively represent a ‘single inventive concept’ as stipulated under Section 10(5) of the Patents Act. This, she contends, fundamentally invalidates the eligibility of the divisional application. Delving deeper, Ms. Dwivedi highlights a pivotal discrepancy: the divisional application revolves around a computing device or system geared to execute the stipulated method. In stark contrast, none of the claims within the parent application even allude to such a computing device or system. She submits that since no system or device claims were initially pursued within the parent application's 183 claims, they should be deemed as abandoned. Moreover, mere disclosure in the complete specification (specifically, features 2100 and 2200) lacks adequate substantiation.

10. To juxtapose the claims and underscore her contention, Ms. Dwivedi draws attention to the following comparative tabulation provided by the Patent Office in their response:

<i>Claims of application Parent Application</i>	<i>Divisional Application</i>	<i>Remarks</i>
<i>Claims 155-162,165-176,179-181 discloses method comprising: computing a motion vector to use in direct mode scaling for a current macroblock in an</i>	<i>Independent Claims 1,6,16,27 discloses that in a computing device that implements a video decoder.</i>	<i>Completely new System Claim in divisional application</i>



<i>interlaced B-field.</i>		
<i>Claims 163-164, 177-178 discloses a computer readable medium storing computer executable instructions</i>	<i>Nowhere in divisional application</i>	

11. Ms. Dwivedi states that the parent application exclusively pertains to method claims relating to computer-readable medium storing executable instructions, whereas, the divisional application focuses on claims about a system or computing device. Consequently, she asserts that there is no demonstration of ‘plurality of inventions’. Therefore, introduction of entirely new claims related to computing devices or systems in the divisional application is not compliant with Section 16 of the Patents Act.

Analysis and findings

12. The Court has heard the counsel for parties. The impugned order cites the following two grounds of rejection:

- a) Claims made in the divisional application are not distinct from the claims of parent application. Thus, the subject invention is not allowable under Section 16 of the Patents Act.
- b) Subject invention is not patentable under Section 3(k) of the Patents Act, being a computer program *per se*.

Qua objection under Section 16

13. The decision-making process: It is noteworthy that the Respondent has rejected the subject invention under Section 16 of the Patents Act without supplying any substantial or coherent justification. Regardless, since the Controller has provided a comprehensive reply outlining their position



on Section 16, this Court finds it prudent to address this ground substantively, eliminating the necessity for any remand on this specific issue.

14. Examination of distinctness of claims: The central inquiry before us revolves around determining if the core of the subject invention, as presented in the divisional application, contravenes Section 16 of the Patents Act. The Controller contends that claims 1 through 29 of this divisional application do not present a unique invention, suggesting that they merely mirror the parent application. To make such a determination, a detailed comparison of the claims from both applications is indispensable. The impugned order, however, seems to harbour an inconsistency. While it points out that the applicant shifted from ‘method’ claims (155-183) to ‘system’ claims (1-29) – a move supposedly not in line with Section 16 – it earlier notes in paragraph 5 as under:

“5. It is evident that the applicant had filed 1-183 claims in the First application (Parent application). There was multiple independent set of claims related to only 'A Method and A Computer readable medium storing computer executable instructions'. There was not even a single claim of 'A computing Device or A System for implementing the method claimed.'”

15. The above-noted observation suggests a glaring omission of any ‘system’ claims in the parent application that aim to operationalize the described method. Intriguingly, the core concept of the invention, initially presented in the method claims of the parent application, re-emerges in the system claims of the divisional application. This latter presentation provides detailed hardware specifications essential for the method’s execution. While there is an evident transition from method to system claims between the parent and divisional applications, it implies a thoughtful reorganization of



the claims. To the Court, however, the exact labelling of these claims – be it as method or system – is of secondary importance.

16. The parent application originally contained 183 claims. Amidst its processing, a conscious choice was made by the Appellant to excise claims 155-183 from this parent application. These claims were subsequently reconstituted into a divisional application with 29 distinct claims.

17. To assess the veracity of the Respondent's contention that the divisional application's claims predominantly echo the claims from the parent application, a comprehensive juxtaposition of the claims of both applications is required. To facilitate this examination, we have encapsulated a side-by-side comparison of the independent claim 1 from both applications in the table below:

<i>Parent Application</i> [487/DELNP/2006]	<i>Divisional Application</i> [8360/DELNP/2010]
A method of decoding encoded video using a video decoder (2200), the method comprising with the video encoder (2200):	A system for implementing a video decoder (2200), the system comprising a processor and memory, wherein the processor is to:
receiving encoded data in a bit stream; and	decode a current direct-mode macroblock in a current interlaced bi-directionally predicted field (2550), including:
decoding an interlaced B-field (2550) using the encoded data, wherein the interlaced B-field (2550) has one or more forward macroblocks and one or more backward macroblocks, and wherein the decoding includes: for each of the one or more forward	determine that a co-located macroblock in a temporally future interlaced predicted field was encoded using four-motion-vector encoding;



macroblocks:	
reconstructing a forward motion vector for the macroblock;	determine one or more motion vectors of the co-located macroblock; and
storing the reconstructed forward motion vector in a corresponding position for the macroblock in a forward motion vector buffer (4510);	calculate a motion vector to use in direct mode scaling operations for the current direct-mode macroblock, including favoring dominant polarity for the one or more motion vectors of the co-located macroblock when calculating the motion vector by: determining whether more of the one or more motion vectors of the co-located macroblock reference a first reference field or a second reference field;
predicting a backward motion vector using one or more motion vector predictor candidates from a backward motion vector buffer (4520) and also using polarity information for the one or more motion vector predictor candidates from the backward motion vector buffer, including selecting between:	if more of the one or more motion vectors of the co-located macroblock reference the second reference field, compute the motion vector to use in direct mode scaling operations using those of the one or more motion vectors of the co-located macroblock that reference the second reference field; and
using, as the predicted backward motion vector, a same polarity backward motion vector predictor that references a first reference field of the interlaced B-field (2550), the first reference field having same polarity as the interlaced B-field (2550), or using, as the predicted backward motion vector, an opposite polarity	otherwise, compute the motion vector to use in direct mode scaling operations using those of the one or more motion vectors of the co-located macroblock that reference the first reference field; and



backward motion vector predictor that references a second reference field of the interlaced B-field (2550), the second reference field having opposite polarity as the interlaced B-field (2550);	
wherein the selecting for the predicted backward motion vector includes counting how many of the one or more motion vector predictor candidates from the backward motion vector buffer are opposite polarity motion vectors and/or counting how many of the one or more motion vector predictor candidates from the backward motion vector buffer are same polarity motion vectors; and	reconstructs the current interlaced bi-directionally predicted field using the decoded current direct-mode macroblock.
storing the predicted backward motion vector in a corresponding position for the macroblock in the backward motion vector buffer (4520); and for each of the one or more backward macro blocks:	
reconstructing a backward motion vector for the macroblock;	
storing the reconstructed backward motion vector in a corresponding position for the macroblock in the backward motion vector buffer (4520);	
predicting a forward motion vector using one or more motion vector predictor candidates from the forward motion vector buffer (4510) and also using polarity information for the one or more motion vector predictor	



candidates from the forward motion vector buffer (4510), including selecting between:	
using, as the predicted forward motion vector, a same polarity forward motion vector predictor that references a third reference field of the interlaced B-field (2550), the third reference field having same polarity as the interlaced B-field (2550), or	
using, as the predicted forward motion vector, an opposite polarity forward motion vector predictor that references a fourth reference field of the interlaced B-field (2550), the fourth reference field having opposite polarity as the interlaced B-field (2550);	
wherein the selecting for the predicted forward motion vector includes counting how many of the one or more motion vector predictor candidates from the forward motion vector buffer (4510) are opposite polarity motion vectors and/or counting how many of the one or more motion vector predictor candidates from the forward motion vector buffer (4510) are same polarity motion vectors and	
storing the predicted forward motion vector in a corresponding position for the macroblock in the forward motion vector buffer (4510).	

18. Claims are pivotal in defining the extent of protection that an



invention receives. Even if the foundational teachings or descriptions appear similar, as illustrated in the afore-noted table, it is the specific framing and content of the claims that truly differentiate one invention from another. The transformation of claim 155 from the parent application into the independent claim 1 of the divisional application underscores this distinction. Consequently, the Controller's stance that the divisional application's claims are merely reiterations of the parent application lacks merit. Furthermore, the Respondent's reliance on the **Boehringer** (*Supra*) judgment is misguided. In that judgment, the Court did not dispute an applicant's right to file a divisional application. Instead, it clarified the circumstances under which such an application would be deemed valid. Given these considerations, deducing that the claims in the divisional application were previously disclosed in the parent application's claims is an erroneous conclusion.

19. The Controller has also failed to appreciate that Section 16 of Patents Act permits voluntary filing of divisional application. In fact, the language of the statute makes it clear that a divisional application can be filed if the claims of the parent application relate to more than one invention. This has also been observed in the case of **Boehringer** (*Supra*), wherein the concept of 'plurality of claims' has been discussed. The true intent behind allowing such filings is to ensure that all potentially patentable inventions, stemming from an original application, get their due protection. It is noteworthy that while there are similarities in the thematic base (both deal with video decoding), the approach taken by each application is distinct. The parent application lends its narrative to the method of video decoding, elucidating steps involved in the process. Conversely, the divisional application is



system-centric, focusing on a structure that implements this video decoding, laying emphasis on the specifics of operations carried out by this system. From the comparison, it is evident that while the subject matter of both applications revolves around video decoding, the *modus operandi* delineated in each is different. One is method-centric, emphasizing on ‘how’ it is done; the other is system-centric, illuminating ‘what’ it does.

20. Furthermore, the Court finds that the Controller has erroneously objected to amendment of method claims 1-29 to systems claims 1-29 under section 16 of the Act. It is crucial to note that we are currently in the pre-grant stage, which is primarily governed by section 59(1) of the Patents Act. Given that the transition from method claims to system claims is encompassed within the complete specifications, such amendments are permissible, as held in ***Allergan INC v. The Controller of Patents***.²

Conclusion

21. From the foregoing analysis, it is evident that the Controller’s conclusions are erroneous. A divisional application by its very nature requires a delineation from its parent application. The table presented earlier, demonstrates a marked transition from method-oriented claims in the parent application to system-centric claims in the divisional one. While the divisional application’s claims draw inspiration from the parent application, they undeniably embody different set of claims. The objections under Section 16 of Patents Act, are therefore not sustainable.

Qua Section 3(k) of Patents Act

² 2023 SCC Online Del 295.



22. Section 3(k) primarily acts as a safeguard against granting patents for abstract ideas or theoretical concepts, which, when related to software, could hinder innovation. The pivotal point here is the term “*per se*”. While a computer program “*per se*” cannot be patented, the legislation does not completely exclude computer-related inventions from patentability. The central question then is whether the subject invention, beyond being just a software code, introduces a novel and non-obvious technical advancement or has a particular industrial application. On this issue, the Controller has merely held as follows:

“Moreover, the claimed matter is a computer program per se stored in a computer readable medium. Hence, the amended claims 1-29 are proscribed under section 3(k) of The Patent Act.”

23. The aforementioned reasoning for disallowing the application under Section 3(k) of the Act is devoid of any reasoning. Ms. Dwivedi has been unable to substantiate this ground during the hearing as well. This Court, on various occasions, held that the reasons for rejection of patent should be clearly discernible from the order, failing which the order is susceptible to judicial intervention.³ As the impugned order fails to meet this threshold, the Court deems it appropriate to remand the matter on this aspect.

Directions

24. For the foregoing reasons, the present appeal is allowed and following directions are issued:

- (i) The impugned order dated 28th November, 2019 is set-aside.
- (ii) As there is no discussion on the requirements under Section 3(k) of

³ See: **Blackberry Limited v. Assistant Controller of Patents and Designs**, order dated 22nd March, 2023 in C.A. (COMM.IPD-PAT) 301/2022 and **Dolby International AB v. The Assistant Controller of Patents**



the Patents Act, the matter is remanded to Respondent for re-examination of the divisional application on the objections pertaining to non-patentability under Section 3(k).

(iii) The divisional application for the subject invention is restored to its original number.

(iv) Prior to deciding the matter afresh, Appellant shall be granted a hearing and the notice of such hearing must clearly delineate the objection(s), if any.

(v) Let the aforesaid exercise be completed and matter be decided within four months from the date of release of this judgement, on the issue of Section 3(k) of the Patents Act. The Appellant's divisional application shall be decided in light of the law discussed in the judgments of this Court in ***Ferid Allani v. Union of India and Ors.***⁴ and ***Microsoft Corporation v. The Assistant Controller of Patents and Designs.***⁵

25. With the above directions, the appeal stands disposed.

SANJEEV NARULA, J

AUGUST 11, 2023

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(Corrected and released on 23rd August, 2023)

and Designs, Neutral Citation No.: 2023/DHC/001854.

⁴ W.P.(C) 7/2014 dated 12th December, 2019.

⁵ DHC Neutral Citation: 2023:DHC:4322.